

Solved selected problems of Modern Physics for Scientists and Engineers - John Taylor

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Chapter 6 - Matter Waves

Solution. 6.2 The wavelength of a golf ball of mass $60g$ with speed $30m/s$ is given by

$$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}}{0.06 \cdot 30} = 3.681 \times 10^{-34} \text{ m}$$

This is a really small amount for a macroscopic object so it's not likely that the wave properties of a golf ball could be easily detected. $\square$

Solution. 6.5 A neutron with the same wavelength as blue light ($\lambda = 450 \text{ nm}$) has a momentum of

$$p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{450 \times 10^{-9}} = 1.473 \times 10^{-27} \text{ kg} \cdot \text{m/s}$$

Hence the kinetic energy is

$$K = \frac{1}{2} \frac{p^2}{m} = \frac{1}{2} \frac{(1.473 \times 10^{-27})^2}{1.6749 \times 10^{-27}} = 6.4771 \times 10^{-28} \text{ J}$$

$\square$

Solution. 6.9 The relativistic energy of an electron is given by

$$E^2 = (pc)^2 + (mc^2)^2$$

Hence if the electron has an energy of $E = 2 \text{ MeV}$ then the momentum is given by

$$p = \frac{1}{c} \sqrt{E^2 - (mc^2)^2} = \frac{1}{c} \sqrt{4 - (0.51)^2} = 1.933 \text{ MeV}/c$$

So the wavelength is given by

$$\begin{aligned} \lambda &= \frac{h}{p} = \frac{4.136 \times 10^{-15} \text{ eV} \cdot \text{s}}{1.933 \times 10^6 \text{ eV}} \cdot 2.998 \times 10^8 \text{ m/s} = 6.414 \times 10^{-13} \text{ m} \\ &= 0,0006414 \text{ nm} \end{aligned}$$

$\square$

Solution. 6.12

- (a) From the relativistic relation for an electron we know that

$$p = \frac{1}{c} \sqrt{E^2 - (mc^2)^2}$$

and for a photon

$$p = \frac{E}{c}$$

Hence from De Broglie, the wavelengths are

$$\lambda_e = \frac{hc}{\sqrt{E^2 - (mc^2)^2}}$$

$$\lambda_{ph} = \frac{hc}{E}$$

So electrons and photons with the same energy E have different wavelengths.

- (b) If $mc^2 \ll E$ then $(mc^2)^2$ is going to be a lot smaller than E hence we can approximate λ_e by dropping the term, hence

$$\lambda_e \approx \frac{hc}{\sqrt{E^2}} = \frac{hc}{E}$$

Which is the same wavelength for a photon.

□

Solution. 6.15 Given 3-eV electrons where $n = 1$ maximum occurs at 15° the slit separation d has to be

$$d = \frac{n\lambda}{\sin \theta}$$

Where λ for a 3 eV electron is

$$\lambda = \frac{hc}{\sqrt{2mc^2K}} = \frac{1240}{\sqrt{2 \cdot 0.51 \times 10^6 \cdot 3}} = 0.708 \text{ nm}$$

Hence

$$d = \frac{0.708}{\sin(15^\circ)} = \frac{0.708}{0.258} = 2.744 \text{ nm}$$

□

Solution. 6.16 Given that the probability of finding an electron in a volume dV at $\mathbf{r}$ is

$$P = |\Psi(\mathbf{r}, t)|^2 dV$$

and P is dimensionless then $\Psi(\mathbf{r}, t)$ must have the dimension of $1/\sqrt{\text{length}^3}$ and hence simplifying the SI units for P gives us

$$\left| \frac{1}{\sqrt{m^3}} \right|^2 m^3 = 1$$

i.e. P is dimensionless as we wanted. □

Solution. 6.19 Let $y(x, t) = A \sin(kx - \omega t)$ be a traveling wave with $A = 4 \text{ cm}$, $k = 12 \text{ rad/cm}$, and $\omega = 2 \times 10^3 \text{ rad/s}$ then the frequency of the wave is

$$f = \frac{\omega}{2\pi} = \frac{2 \times 10^3}{2\pi} = 318.31 \text{ Hz}$$

The wavelength is

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{12} = 0.52 \text{ cm}$$

And hence the velocity of the wave is

$$v = \lambda f = 0.52 \cdot 318.31 = 165.52 \text{ cm/s}$$

□

Solution. 6.20 The SI units of the wave parameters are

$$\lambda : [m]$$

$$T : [s]$$

$$f : [s^{-1}] = [Hz]$$

$$\omega : [rad/s]$$

$$v : [m/s]$$

□

Solution. 6.22 For X-rays with $\lambda = 0.05 \text{ nm}$, k is given by

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{0.05 \times 10^{-9}} = 125.66 \times 10^9 \text{ 1/m}$$

but since X-rays have speed c then

$$f = \frac{c}{\lambda} = \frac{2.998 \times 10^8}{125.66 \times 10^9} = 0.00238 \text{ Hz}$$

And hence ω is

$$\omega = 2\pi f = 2\pi \cdot 0.00238 = 0.0149 \text{ 1/s}$$

□

Solution. 6.23 For a non-relativistic electron we know that

$$E = \frac{p^2}{2m}$$

Hence p for a 300 eV electron is

$$p = \sqrt{2mE} = \sqrt{2 \cdot 0.511 \times 10^6 \cdot 300} = 17509.99 \text{ eV}/c$$

Now, by using de Broglie relation we determine λ as follows

$$\begin{aligned} \lambda &= \frac{h}{p} \\ &= \frac{4.136 \times 10^{-15} \text{ eVs}}{17509.99 \text{ eV}} \cdot 2.998 \times 10^8 \text{ m/s} \\ &= 7.081 \times 10^{-11} \text{ m} \end{aligned}$$

Then k is

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{7.081 \times 10^{-11}} = 88733022273.40 \text{ m}^{-1}$$

□

Solution. 6.26

- (a) Let us consider the wave $y(x, t) = A \sin(2\pi(x/\lambda - t/T))$. Let us focus attention on the wave crest for which the argument of the sine function is $\pi/2$ then must be that

$$\begin{aligned} 2\pi \left(\frac{x}{\lambda} - \frac{t}{T} \right) &= \frac{\pi}{2} \\ x &= \lambda \left(\frac{t}{T} + \frac{1}{4} \right) \end{aligned}$$

Then taking the derivative of x with respect to time we get that the velocity is

$$v = \frac{dx}{dt} = \frac{\lambda}{T}$$

- (b) If the minus sign on the wave equation is replaced by a plus sign, doing the same analysis we get that

$$\begin{aligned} 2\pi \left(\frac{x}{\lambda} + \frac{t}{T} \right) &= \frac{\pi}{2} \\ x &= \lambda \left(\frac{1}{4} - \frac{t}{T} \right) \end{aligned}$$

And hence the time derivative is

$$v = \frac{dx}{dt} = -\frac{\lambda}{T}$$

This implies that the wave moves in the opposite direction i.e. to the left.

□

Solution. 6.27 Let a wave $\Psi = A \sin(ax - bt)$. The crests of the wave happens when the argument of the sine function is $\pi/2 + 2\pi n$ where $n \in \{0\} \cup \mathbb{N}$ this implies that

$$ax - bt = \frac{\pi}{2} + 2\pi n$$

$$x = \frac{1}{a} \left(\frac{\pi}{2} + 2\pi n + bt \right)$$

Then taking the derivative of x with respect to time we get that the velocity is

$$v = \frac{dx}{dt} = \frac{b}{a}$$

□

Solution. 6.30

- (a) By inspection, we see that the wave has a period approximately of 5 ms so the fundamental frequency is $f = 1/5 \times 10^{-3} = 200 \text{ Hz}$.
- (b) By inspection, we see that the highest harmonic is the number 4 since we see a wave added to the fundamental with a wavelength 4 times shorter than the fundamental i.e. with 4 periods inside the fundamental period. Its frequency is $f_4 = 1/1.25 \times 10^{-3} = 800 \text{ Hz}$

□

Solution. 6.32 From equation (6.58) we know that

$$A_0 = \frac{1}{\lambda} \int_0^\lambda F(x) dx$$

Then for the case of a periodic square pulse of amplitude 1 this equation becomes

$$\begin{aligned} A_0 &= \frac{1}{\lambda} \left[\int_0^{a/2} 1 \, dx + 0 + \int_0^{a/2} 1 \, dx \right] \\ &= \frac{1}{\lambda} \left[\frac{a}{2} + \frac{a}{2} \right] \\ &= \frac{a}{\lambda} \end{aligned}$$

On the other hand, from equation (6.59) we know that

$$A_n = \frac{2}{\lambda} \int_0^\lambda F(x) \cos\left(\frac{2n\pi x}{\lambda}\right) dx$$

Then for this case we have that

$$\begin{aligned} A_n &= \frac{2}{\lambda} \left[\int_0^{a/2} \cos\left(\frac{2n\pi x}{\lambda}\right) dx + 0 + \int_0^{a/2} \cos\left(\frac{2n\pi x}{\lambda}\right) dx \right] \\ &= \frac{2}{\lambda} \left[\frac{\lambda}{2n\pi} \sin\left(\frac{n\pi a}{\lambda}\right) + \frac{\lambda}{2n\pi} \sin\left(\frac{n\pi a}{\lambda}\right) \right] \\ &= \frac{2}{n\pi} \sin\left(\frac{n\pi a}{\lambda}\right) \end{aligned}$$

□

Solution. 6.33 To determine Δx for the rectangular wave we need to determine first $P(x)$ but since the height of the pulse shown must be $1/2a$ so the total probability of finding a particle anywhere is 1 then we have that the rms spread becomes

$$\begin{aligned}
 \Delta x &= \sqrt{\int_{-\infty}^{-a} 0 \, dx + \int_{-a}^a x^2 \left(\frac{a}{2}\right) \, dx + \int_a^{\infty} 0 \, dx} \\
 &= \sqrt{\frac{a}{2} \int_{-a}^a x^2 \, dx} \\
 &= \sqrt{\frac{a}{2} \left[\frac{x^3}{3} \right]_{-a}^a} \\
 &= \sqrt{\frac{a^4}{3}} \\
 &= \frac{a^2}{\sqrt{3}}
 \end{aligned}$$

□

Solution. 6.35

(a) Let us integrate equation (6.57) from 0 to λ then

$$\begin{aligned}
\int_0^\lambda F(x) dx &= \int_0^\lambda \sum_{n=0}^{\infty} A_n \cos\left(\frac{2n\pi x}{\lambda}\right) dx \\
&= \int_0^\lambda A_0 dx + \sum_{n=1}^{\infty} \int_0^\lambda A_n \cos\left(\frac{2n\pi x}{\lambda}\right) dx \\
&= A_0\lambda + \sum_{n=1}^{\infty} \left[\frac{A_n\lambda}{2n\pi} \sin\left(\frac{2n\pi x}{\lambda}\right) \right]_0^\lambda \\
&= A_0\lambda + \sum_{n=1}^{\infty} \frac{A_n\lambda}{2n\pi} [\sin(2n\pi) - \sin(0)] \\
&= A_0\lambda
\end{aligned}$$

Where we used that $\sin(2n\pi) = 0$ for $n \geq 0$. Therefore we have that

$$A_0 = \frac{1}{\lambda} \int_0^\lambda F(x) dx$$

(b) Let us multiply equation (6.57) by $\cos(2m\pi x/\lambda)$ and also integrate both sides from 0 to λ then

$$\int_0^\lambda F(x) \cos\left(\frac{2m\pi x}{\lambda}\right) dx = \sum_{n=0}^{\infty} \int_0^\lambda A_n \cos\left(\frac{2n\pi x}{\lambda}\right) \cos\left(\frac{2m\pi x}{\lambda}\right) dx$$

According to Appendix B $\int_0^\lambda \cos\left(\frac{2n\pi x}{\lambda}\right) \cos\left(\frac{2m\pi x}{\lambda}\right) dx$ is 0 if $m \neq n$ and $\lambda/2$ if $m = n$ so we get that

$$\begin{aligned}
\int_0^\lambda F(x) \cos\left(\frac{2m\pi x}{\lambda}\right) dx &= A_m \int_0^\lambda \cos^2\left(\frac{2m\pi x}{\lambda}\right) dx \\
&= A_m \left[\frac{\lambda \sin\left(\frac{4m\pi x}{\lambda}\right)}{8m\pi} + \frac{x}{2} \right]_0^\lambda \\
&= A_m \frac{\lambda}{2}
\end{aligned}$$

Where we used that $\sin(4m\pi) = 0$ for $m \geq 0$. Therefore we have that

$$A_m = \frac{2}{\lambda} \int_0^\lambda F(x) \cos\left(\frac{2m\pi x}{\lambda}\right) dx$$

□

Solution. 6.36

(a) The Fourier expansion of an odd function is given by

$$F(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{2n\pi x}{\lambda}\right)$$

Then multiplying this equation by $\sin(2m\pi x/\lambda)$ and integrating both sides from 0 to λ we get that

$$\int_0^{\lambda} F(x) \sin\left(\frac{2m\pi x}{\lambda}\right) dx = \sum_{n=1}^{\infty} B_n \int_0^{\lambda} \sin\left(\frac{2n\pi x}{\lambda}\right) \sin\left(\frac{2m\pi x}{\lambda}\right) dx$$

But according to Appendix B $\int_0^{\lambda} \sin\left(\frac{2n\pi x}{\lambda}\right) \sin\left(\frac{2m\pi x}{\lambda}\right) dx$ is 0 if $m \neq n$ and $\lambda/2$ if $m = n$ then

$$\int_0^{\lambda} F(x) \sin\left(\frac{2m\pi x}{\lambda}\right) dx = B_m \int_0^{\lambda} \sin^2\left(\frac{2m\pi x}{\lambda}\right) dx = B_m \frac{\lambda}{2}$$

Therefore

$$B_m = \frac{2}{\lambda} \int_0^{\lambda} F(x) \sin\left(\frac{2m\pi x}{\lambda}\right) dx$$

(b) For the sawtooth function if we take $\lambda = 1$ then $F(x)$ is defined as

$$F(x) = \begin{cases} 2x & \text{for } 0 \leq x \leq 1/2 \\ -2x + 2 & \text{for } 1/2 \leq x \leq 1 \end{cases}$$

Then the equation for Fourier coefficients become

$$\begin{aligned} B_n &= 2 \int_0^1 F(x) \sin(2n\pi x) dx \\ &= 4 \left[\int_0^{1/2} x \sin(n\pi x) dx + \int_{1/2}^1 (1-x) \sin(n\pi x) dx \right] \\ &= 4 \left[\left[\frac{\sin(n\pi x)}{n^2\pi^2} - \frac{x \cos(n\pi x)}{n\pi} \right]_0^{1/2} + \left[\frac{(x-1) \cos(n\pi x)}{n\pi} - \frac{\sin(n\pi x)}{n^2\pi^2} \right]_{1/2}^1 \right] \\ &= 4 \left[\left[\frac{\sin(n\pi/2)}{n^2\pi^2} - \frac{\cos(n\pi/2)}{2n\pi} \right] + \left[-\frac{\sin(n\pi)}{n^2\pi^2} + \frac{\cos(n\pi/2)}{2n\pi} + \frac{\sin(n\pi/2)}{n^2\pi^2} \right] \right] \\ &= 4 \left[\frac{2 \sin(n\pi/2)}{n^2\pi^2} - \frac{\sin(n\pi)}{n^2\pi^2} \right] \\ &= \frac{32 \sin^3(n\pi/4) \cos(n\pi/4)}{n^2\pi^2} \end{aligned}$$

If n is even we have that $B_n = 0$ since $\sin(n\pi/2) = \sin(n\pi) = 0$ but if n is odd we see that

$$\begin{aligned} B_1 &= \frac{32 \sin^3(\pi/4) \cos(\pi/4)}{\pi^2} = \frac{8}{\pi^2} \\ B_3 &= \frac{32 \sin^3(3\pi/4) \cos(3\pi/4)}{3^2 \pi^2} = -\frac{8}{3^2 \pi^2} \\ B_5 &= \frac{32 \sin^3(5\pi/4) \cos(5\pi/4)}{5^2 \pi^2} = \frac{8}{5^2 \pi^2} \\ B_7 &= \frac{32 \sin^3(7\pi/4) \cos(7\pi/4)}{7^2 \pi^2} = -\frac{8}{7^2 \pi^2} \end{aligned}$$

Then we can say that

$$B_n = \frac{8(-1)^{(n-1)/2}}{n^2 \pi^2}$$

□

Solution. 6.37 The proton is known to be within an interval of $\pm 6 \text{ fm}$ then the spread in the position is certainly no larger than 6 fm i.e.

$$\Delta x \leq 6 \text{ fm}$$

According to the uncertainty relation, this implies that the momentum is uncertain by an amount

$$\Delta p \geq \frac{\hbar}{2\Delta x} = \frac{1.054 \times 10^{-34} \text{ Js}}{1.2 \times 10^{-14} \text{ m}} = 8.78 \times 10^{-21} \text{ kg m/s}$$

Therefore, the minimum uncertainty in its velocity is

$$\Delta v = \frac{\Delta p}{m} \geq \frac{8.78 \times 10^{-21}}{1.672 \times 10^{-27}} = 5251196.17 \text{ m/s}$$

Or as a fraction of c we get that

$$5251196.17 \cdot \frac{c}{2.998 \times 10^8} = 0.017c$$

□

Solution. 6.39 The position of a 60 g golf ball sitting on a tee is determined within $\pm 1 \mu m$ then the spread in the position is certainly no larger than $1 \mu m$ i.e.

$$\Delta x \leq 1 \mu m$$

Then the momentum is uncertain by

$$\Delta p \geq \frac{\hbar}{2\Delta x} \geq \frac{1.054 \times 10^{-34} Js}{2 \times 10^{-6} m} = 5.27 \times 10^{-29} kg m/s$$

Then the minimum possible energy is given by

$$\langle K \rangle \geq \frac{(\Delta p)^2}{2m} = \frac{(5.27 \times 10^{-29})^2}{2 \cdot 0.06} = 2.31 \times 10^{-56} J$$

The corresponding velocity associated to this kinetic energy is uncertain by

$$\Delta v = \frac{\Delta p}{m} \geq \frac{5.27 \times 10^{-29}}{0.06} = 8.78 \times 10^{-28} m/s$$

Hence the ball will move in a year

$$8.78 \times 10^{-28} m/s \cdot 31536000 s = 2.77 \times 10^{-20} m$$

□

Solution. 6.44

- (a) Using the principle of conservation of energy if we assume that the body scaping will be at rest at infinity, we can write that

$$\frac{1}{2}mv^2 - \frac{GmM}{R} = 0$$

Hence the velocity the body should have to scape is

$$v = \sqrt{\frac{2GM}{R}}$$

- (b) For a black hole v_{esc} is computed taking the radius Δx so the energy density is large enough, then by equating this to c we get that

$$\begin{aligned} c^2 &= \frac{2GM}{\Delta x} \\ c^4 &= \frac{2GMc^2}{\Delta x} \\ c^4 &= \frac{2GE}{\Delta x} \end{aligned}$$

Where we multiplied by c^2 and we used that $E = mc^2$. But also for a photon we know that $E = \Delta pc$ hence

$$\begin{aligned} c^4 &= \frac{2G\Delta pc}{R_s} \\ c^3 &= \frac{G\hbar}{\Delta x^2} \\ \Delta x &= \sqrt{\frac{G\hbar}{c^3}} \end{aligned}$$

Where we used that $\Delta p = \hbar/2\Delta x$ assuming Δx is the smallest radius the black hole can have.

- (c) The value of the Planck length is then

$$\Delta x = \sqrt{\frac{G\hbar}{c^3}} = \sqrt{\frac{6.67 \times 10^{-11} \cdot 1.054 \times 10^{-34}}{(2.998 \times 10^8)^3}} = 1.615 \times 10^{-35} \text{ m}$$

- (d) The diameter of a proton in meters is 1.534×10^{-18} so in Planck lengths the diameter of a proton is 9.498×10^{16} .

□

Solution. 6.46 Given that the particle's mean life is of the order 10^{-23} s then the minimum approximate uncertainty in its energy is

$$\Delta E \approx \frac{\hbar}{2\Delta t} = \frac{6.582 \times 10^{-16} \text{ eVs}}{2 \cdot 10^{-23} \text{ s}} = 32.91 \text{ MeV}$$

□

Solution. 6.48

- (a) Given that the particle's lifetime is 1 ms then the minimum approximate uncertainty in its energy is

$$\Delta E \approx \frac{\hbar}{2\Delta t} = \frac{6.582 \times 10^{-16} \text{ eVs}}{2 \cdot 10^{-3} \text{ s}} = 3.291 \times 10^{-13} \text{ eV}$$

- (b) We know that the photon will have an energy uncertainty of $\Delta E = h\Delta f$ but for a photon we also have that $\Delta f = c/\Delta\lambda$ then the uncertainty in wavelength is given by

$$\Delta\lambda = \frac{hc}{\Delta E} = \frac{1240 \text{ eVnm}}{3.291 \times 10^{-13} \text{ eV}} = 3.767 \times 10^{15} \text{ nm}$$

Then the fractional uncertainty is

$$\frac{\Delta\lambda}{\lambda} = \frac{3.767 \times 10^{15} \text{ nm}}{550 \text{ nm}} = 6.849 \times 10^{12}$$

□

Solution. 6.49 For a relativistic particle we know by the Pythagorean relation that

$$E = \sqrt{(m_0c^2)^2 + (pc)^2}$$

Where m_0 is the rest mass. If we consider that m is the relativistic mass defined as

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

Then we can write that the velocity of a relativistic free particle is

$$v_{part} = p/m$$

Also, by equation (6.56) we know that $v_{pack} = dE/dp$ therefore

$$v_{pack} = \frac{d}{dp} \left(\sqrt{(m_0c^2)^2 + (pc)^2} \right) = \frac{pc^2}{\sqrt{(m_0c^2)^2 + (pc)^2}} = \frac{pc^2}{E} = \frac{pc^2}{mc^2} = \frac{p}{m}$$

Therefore even for relativistic particles we have that $v_{part} = v_{pack}$. □